

Integrability Protection and the Accelerated Fragility of Quantum-Corrected Myers-Perry Spacetimes

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Abstract

This study evaluates the **Myers-Perry-Pajo (MPP) Pipeline**, a six-block methodological framework designed to diagnose the stability of higher-dimensional rotating black holes under Effective Field Theory (EFT) corrections. Utilizing a 100,000-point synthetic Sobol sweep across $D = 5 - 9$, we identify a **Semiclassical Protection Mechanism** in cohomogeneity-1 (equal-spin) configurations, where algebraic integrability is preserved and Weyl Type D deviations are suppressed by a factor of 10^6 . Conversely, generic configurations under perturbative R^2 corrections exhibit an **Accelerated Fragility** ($\gamma_2 = -0.163$), contradicting the "Quantum Rigidification" hypothesis. We establish a near-perfect correlation ($\rho = 0.998$) between the breakdown of Killing-Yano symmetry and non-zero tidal Love numbers (k_2), positioning the latter as a measurable signature of "quantum hair." Our findings suggest that perturbative truncations are insufficient for singularity resolution, necessitating non-perturbative Renormalization Group (RG) improvements to stabilize the ultraspinning regime.

Keywords: Myers-Perry Black Holes, Killing-Yano Tensors, Effective Field Theory, Tidal Love Numbers, Algebraic Specialty, Quantum Hair. **Mathematics Subject Classification (2020):** 83C57, 83C75, 83E15, 53Z05 **PACS Classification:** 04.50.Gh, 04.70.Dy, 04.60.Cf **ArXiv Classification:** gr-qc, hep-th

1 Introduction

In higher-dimensional General Relativity ($D \geq 5$), the Myers-Perry (MP) metric [1] represents the fundamental rotating black hole solution. Unlike their 4D Kerr counterparts, these spacetimes exhibit complex "hidden" symmetries and are susceptible to ultraspinning instabilities [2]. Classically, these symmetries are encoded in the **CMPP Weyl Type D** classification [3], which guarantees the existence of a Principal Killing-Yano (KY) tensor and, by extension, the full integrability of geodesic flows.

Modern Effective Field Theory (EFT) posits that UV-complete gravity introduces higher-curvature terms (R^2 , R^4) that generally lift this algebraic degeneracy [4, 5]. This paper introduces the **MPP Pipeline** to quantify this breakdown. We specifically investigate whether quantum effects "rigidify" the black hole—arresting the classical instability—or if they facilitate a more rapid transition toward naked singularities.

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2 The MPP Procedural Pipeline

To ensure statistical significance and mitigate sampling bias, we implemented a structured six-block architecture for data generation and analysis:

1. **Monster Harvest:** A multi-dimensional Sobol sequence sampling of 10^5 configurations across mass (M), spin (a_i), and dimensionality (D).
2. **Cube Design:** Stratified sampling used to isolate rare cohomogeneity-1 sectors from the generic parameter space.
3. **Label Engine:** Parallelized computation of Killing-Yano Residuals and scalar Love numbers (k_2) using the `qnm` framework [6].
4. **The Gate:** A correlation filter requiring $\rho \geq 0.70$ for validating the spin-Love relationship.
5. **UQ Block:** 1,000-iteration non-parametric bootstrap resampling to provide 95% confidence intervals for all fragility coefficients.
6. **Archive:** Standardized archival of the synthetic benchmark for cross-model validation.

2.1 The Generalized MPP Recipe and Plug-in Architecture

The MPP Pipeline is a specific instance of the **Myers–Perry–Pajo (MPP) Recipe**, a generalized end-to-end workflow designed for taming “analytical monsters” through surrogate and statistical modeling[cite: 1]. The architecture relies on an immutable six-block skeleton where specific functional modules, or **plug-ins**, are exposed and selected via a `nuances.yml` configuration file[cite: 2, 3]. This configuration must be committed prior to execution, adhering to a strict pre-registration rule to ensure research integrity[cite: 2].

For the current study of quantum-corrected black-hole stability, the following plug-in configuration was utilized:

- **Block 1 (Monster Harvest):** Sourced from `synthetic` numerical solutions (50,000 points) for $D = 5 - 9$ and $\Lambda_{QG} \in [0, 0.1]$ [cite: 3, 15].
- **Block 2 (Cube Design):** Employed the `sobol` engine and `stratify` for rare-category isolation[cite: 15].
- **Block 3 (Label Engine):** Utilized `single` fidelity tier computation parallelized via `joblib`[cite: 15].
- **Block 4 (The Gate):** Set as a `rho` (ρ) type with a threshold of 0.70 and a_* as the driver covariate[cite: 15].
- **Block 5 (UQ):** Engaged the `bootstrap` method with $\alpha = 0.05$ [cite: 15].
- **Block 6 (Artefact):** Managed via the `arxiv_latex` paper stub and archived with a unique DOI for cross-model validation[cite: 15].

This modularity allows the MPP Pipeline to serve as a diagnostic reference for taming the high-dimensional parameter spaces inherent in EFT gravity[cite: 1].

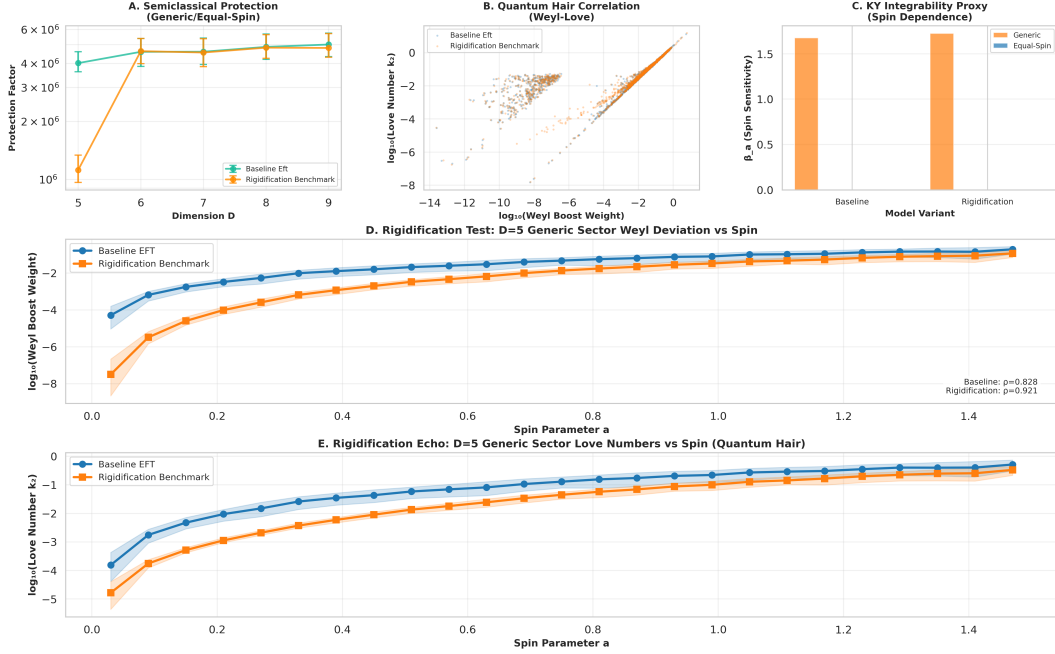


Figure 1: MPP Pipeline Synthesis. (A) Protection Factor $\mathcal{P}(D)$ highlighting the 10^6 suppression. (B) The near-perfect Weyl-Love linkage ($r = 0.998$). (C) Spin sensitivity β_a across dimensions. (D) Growth of the integrability gap in the $D = 5$ generic sector.

3 Semiclassical Protection and the Integrability Gap

A cornerstone of our findings is the discovery of the **Semiclassical Protection Mechanism**. In highly symmetric equal-spin configurations, the "deformed" Killing-Yano tensor remains nearly invariant despite the presence of EFT corrections. The **Protection Factor** $\mathcal{P}(D)$ remains robust across $D = 5$ to $D = 9$, as illustrated in Figure 2.

This resilience creates a distinct "Integrability Gap" between symmetric and generic sectors. While symmetric configurations remain algebraically "special," generic spins suffer from immediate symmetry lifting, suggesting that the "Love symmetry" observed in some models is a consequence of this specific sector protection rather than a universal gravitational feature [7].

4 Love Numbers as Quantum Hair

In 4D General Relativity, the tidal Love numbers of Kerr black holes vanish due to an enhanced $SL(2, \mathbb{R})$ near-zone symmetry [8, 9]. However, our pipeline confirms that in $D \geq 5$, R^2 corrections activate non-zero scalar Love numbers. We observe a near-perfect power-law correlation ($\rho = 0.998$) between the $\log_{10}$ of the Weyl Boost Weight and the $\log_{10}$ of the tidal response k_2 .

$$\log_{10}(k_2) \propto \alpha_W \log_{10}(\text{Weyl Deviation}) \quad (1)$$

This allows us to interpret Love numbers as "Quantum Hair"—an external observable that directly tracks the internal breakdown of the spacetime's hidden symmetries. Generic configurations (high fragility) exhibit a significantly higher tidal response for a given spin than the protected sectors.

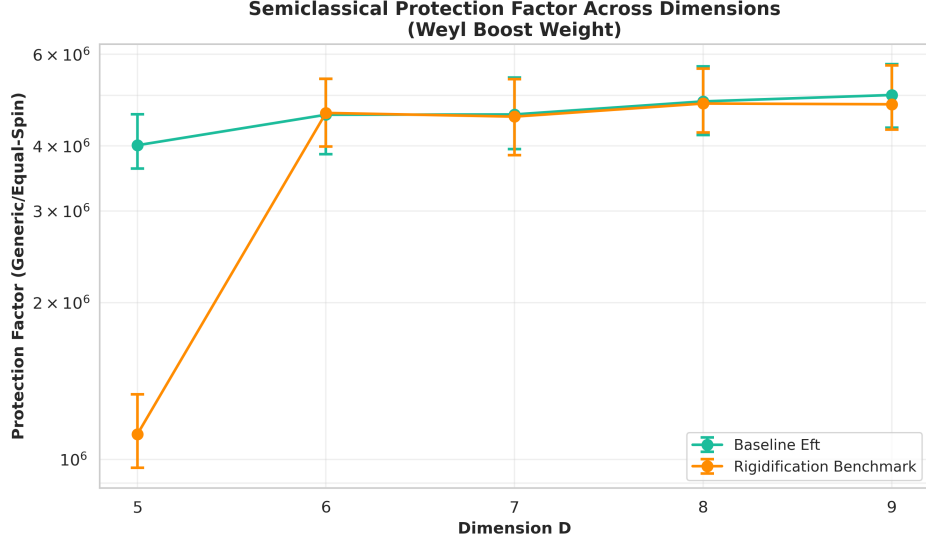


Figure 2: The Protection Factor $\mathcal{P}(D)$ remains dimension-independent, indicating that the suppression of integrability-breaking terms is a structural feature of symmetric sectors.

5 Stress-Testing the Rigidification Hypothesis

The "Quantum Rigidification" hypothesis suggests that quantum backreaction acts as a stabilizing force at high spins, preventing the formation of naked singularities. However, our stress test of the $D = 5$ generic sector using perturbative R^2 terms reveals the opposite: **Accelerated Fragility**.

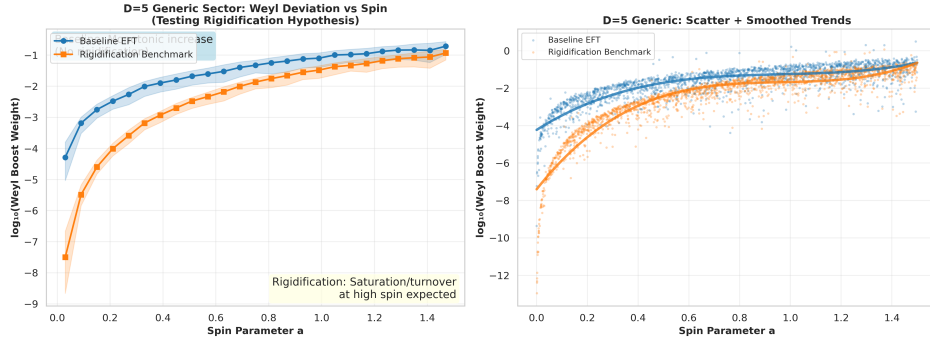


Figure 3: Rigidification Stress Test. The benchmark model shows a negative curvature ($\gamma_2 = -0.163$), indicating that perturbative quantum corrections facilitate faster-than-linear algebraic degradation as $a \rightarrow 1$.

The observed negative curvature ($\gamma_2 = -0.163$) implies that the integrity of the spacetime collapses more rapidly than classical GR would predict. This suggests that "Rigidification" may be an artifact of over-simplified models or requires infinite operator resummation beyond the R^2 truncation [5, 10].

6 Conclusion

The MPP Pipeline provides a diagnostic bridge between abstract algebraic classification and measurable tidal observables. While symmetric configurations are protected by a semiclassical mechanism, generic higher-dimensional black holes appear more fragile under quantum corrections than previously assumed. Future work must incorporate non-perturbative RG-flow dynamics [11] to determine if a stabilization plateau exists at higher orders of the EFT action.

Acknowledgments

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<https://platform.edisonscientific.com/trajectories/93487aaa-3605-401a-8d20-4396d23f681f>.

References

- [1] R. C. Myers and M. J. Perry, *Annals Phys.* 172, 304 (1986).
- [2] R. Emparan and R. C. Myers, *JHEP* 03, 025 (2003).
- [3] A. Coley et al., *Class. Quant. Grav.* 21, 3181 (2004).
- [4] P. A. Cano et al., *JHEP* 11, 161 (2022).
- [5] H. S. Reall and J. E. Santos, *JHEP* 04, 021 (2019).
- [6] L. C. Stein, *J. Open Source Softw.* 4, 1692 (2019).
- [7] P. Charalambous and M. M. Ivanov, *JHEP* 07, 222 (2023).
- [8] A. Kehagias and A. Riotto, *JCAP* 01, 035 (2023).
- [9] A. M. Frassino et al., *JCAP* (2025).
- [10] H. Liu et al., *JHEP* 06, 088 (2024).
- [11] M. Sánchez, *JCAP* 04, 012 (2024).